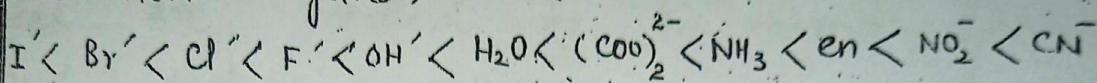


2: SPECTROCHEMICAL SERIES:- By drawing the visible spectra of complexes, it is possible to arrange ligands in the order of increasing crystal field strengths. Some common ligands are



Such an arrangement is known as

"Spectrochemical Series."

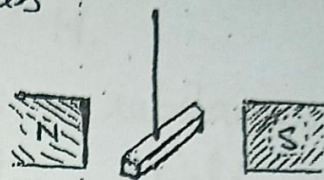
- (i) Strong field ligands like CN^- usually produce low spin complexes where possible and high crystal field splitting.
- (ii) Weak field ligands such as Cl^- usually produce high spin complexes and small crystal field splitting.

It is seen that low spin complexes usually absorb high energy (shorter λ) light than high spin complexes.

DIAMAGNETISM AND PARAMAGNETISM

DIAMAGNETISM: The substances which when placed in magnetic field, decrease the intensity of magnetic field than in the vacuum are called diamagnetic substances and that property due to which they show this

behaviour is called diamagnetism. The magnetic lines of force tend to avoid such substances and as such diamagnetic substances are repelled by the magnetic field and such substances set themselves at right angle to the magnetic field as in fig.



Origion of Diamagnetism: If two electrons (with opposite spins are paired in the same orbital, the magnetic field produced by one electron is cancelled by that caused by the other electron, because each of the two electron has equal and opposite magnetic moment. Thus the substance having only paired electrons give zero resultant magnetic moment and consequently are diamagnetic.

Diamagnetism is temperature-independent and is shown by all types of substances (even by paramagnetic substances). Since diamagnetism is much weaker than Paramagnetism (1 to 100 times weaker) and both act opposite to each other, it is difficult for the

(1)

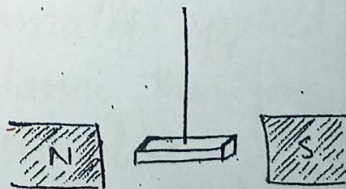
substances having paired electrons to show diamagnetism i.e. the substances having unpaired electrons show a net paramagnetism.

For diamagnetic substances $B < H$. As μ is less than 1, χ will be negative. Hence I is opposing H .

PARAMAGNETISM :- The substances which when placed in a magnetic field, allow the magnetic lines of force to pass through them rather than through vacuum are called paramagnetic substances and the property due to which they show this behaviour is called paramagnetism. A paramagnetic substance tends to set itself with its length parallel to the magnetic field. Thus a paramagnetic substance is attracted into a magnetic field, as in fig.

Paramagnetism is sub-divided into

- (i) Normal Paramagnetism
- (ii) Temperature independent or Van Vleck paramagnetism (TIP)
- (iii) Pauli or Free electron paramagnetism.



For paramagnetic substances $B > H$. As μ is greater than 1, χ will be +ve. Hence I is acting in direction of H .

Origin of Paramagnetism: Paramagnetism of a substance consisting of atoms, ions or molecules is caused by the presence of unpaired electrons in the substance. The greater the no. of unpaired electrons, the greater will be paramagnetism shown by the substance.

In substances containing one or more unpaired electrons the magnetic field caused by these unpaired electrons are not mutually cancelled, since each of unpaired electrons has equal magnetic moment and thus some permanent and definite values of resultant magnetic moment is obtained. This resultant magnetic moment which is a combination of spin and orbital magnetic moments is sufficiently of high magnitude to overcome small magnetic moments induced by the externally applied magnetic field. Such a substance, therefore instead of experiencing repulsion like diamagnetic substance, experiences attraction in a magnetic field i.e. it shows paramagnetic behaviour.

Paramagnetic molecules are free to orient themselves ideally in the gas phase.

but in practise also in solution and even in the solid state, if magnetically dilute. They are subject to two opposing effects

- i) Magnetic field H which tends to align the molecular magnets in the same direction as that of the field
- ii) Thermal agitation or thermal energy KT (called KT effect) which tends to orient molecular ~~moments~~ magnets randomly. The effect of applied magnetic field becomes more pronounced with the decrease in the value of absolute temperature T .

“JOHN-TELLER THEOREM”

This theorem states that any non-linear molecular system possessing degenerate electronic state is unstable and a distortion must occur to form a system of lower symmetry and lower energy. This formation will remove degeneracy and increase stability.

1: This theorem was put forward by John-Teller in 1937 to explain why certain 6 coordinated system undergo distortion.

"Magnetism"

If an electric current which is a flow of e^- is allowed to flow through a wire coiled around a metal core, a field is produced.

According to classical model of an atom, the e^- has two types of motions

(i) Orbital Motion:

which is due to the motion of the e^- round the nucleus of atom in an orbit. Orbital motion can be compared to the flow of electric current through a coiled wire. The orbital motion produces magnetic field or magnetic moment which is called "Orbital magnetic moment".

(ii) Spin motion:

which is due to the spinning of e^- round its own axis. This spin motion produces magnetic field or magnetic moment which is called "Spin magnetic moment".

These magnetic moments make an atom behave like a small magnet, i.e. it is these two magnetic moments which produce magnetic properties in substance.

Magnetic behaviour:

When a substance is placed b/w

the poles of magnet, the magnetic field produce by the orbital and the spin motion of the e^- s interacts externally applied magnetic field. So the various substance show different behaviours which are known "magnetic behaviour".

These magnetic behaviour are classified as

- (i) Diamagnetism, (ii) paramagnetism,
- (iii) ferromagnetism (iv) Antiferromagnetism
- (v) ferrimagnetism.

Last three are rare occurrence but Diamagnetism and paramagnetism are of great importance.

Magnetically dilute substance:

In which the individual magnetic centers in the crystal lattice are apart to minimise any further magnetic interaction b/w near neighbour.

Magnetically concentrated substance:

In which individual magnetic centres in the crystal lattices are close enough to ϕ initiate the further magnetic interaction called magnetically concentrated substance.

Diamagnetism:-

The substance, when placed in a magnetic field, decrease the intensity of the magnetic field then in vacuum and called diamagnetic substance and the property due to which they show this behaviour is called diamagnetism.

Such substance are repelled by magnetic field.

Origin of Diamagnetism:

Thus the substance having only paired e^- give zero resultant magnetic moment.

Diamagnetism is much weaker than paramagnetism (1 to 100 times weaker) and both act opposite to each other. It is difficult for the substance having unpaired e^- to show diamagnetism.

Paramagnetism:

The substance which, when placed in a magnetic field, allow the magnetic lines of force to pass through them rather than through vacuum.

are called paramagnetic substance and the property due to which they show behaviour is called paramagnetism.

Origin of Paramagnetism:

The paramagnetism is caused by the presence of unpaired e^- s in the substance. The magnetic fields caused by these unpaired e^- s are not mutually cancelled. Such a substance therefore, instead of experiencing repulsion like diamagnetic substance, experience attraction in a magnetic field i.e. it shows paramagnetic behaviour.

Preparation of Coordination Compd:

(i) Substitution Reactions in aqueous Soln:

The substitution reaction in aqueous soln is by far the most common method used for the synthesis of metal complexes. The method involves a reaction b/w a metal salt in water soln and a coordinating agent.

Example:

The complex $[Cu(NH_3)_4]SO_4$ prepared by reaction b/w an aqueous solution of $CuSO_4$ and excess NH_3 . The coordinated H_2O is replaced by ammonia which is indicated by the change in color from light blue to dark blue.

- (v) Since diamagnetic substance has all the electrons in the paired state or has no unpaired electrons, μ_{eff} value of diamagnetic substances is zero ($\mu_{\text{eff}} = 0$).
- (vi) Diamagnetism is a property of completely - filled electronic subshells. Since all elements except hydrogen atom contain filled electronic shells, some diamagnetism is shown by all substances. The presence of diamagnetism in all substances is due to the interaction of the magnetic field with the paired electrons present in shells which oppose the field. Since the paramagnetic substances also have some amount of diamagnetic character, the magnetic susceptibility of paramagnetic substances (χ_{para}) should be corrected for diamagnetic susceptibility. This can be done with the help of Pascals constants. The magnitude of diamagnetism present in substances is about 100 times less than the paramagnetism.
- (vii) The ions of the metals of 2nd and 3rd transition series are diamagnetic, if the ions have even number of electrons.

Ferromagnetism

Substances which show permanent magnetism even in the absence of the magnetic field are called ferromagnetic substances. Such substances remain permanently magnetised, once they have been magnetised. This type of magnetism arises due to spontaneous alignment of magnetic moments due to unpaired electrons in the same direction, as shown in Figure 25.10. This is a case of large amount of paramagnetism.

Ferromagnetism is shown by the substances which contain atoms or ions with incomplet *d* or *f* subshells. Examples of substances which show ferromagnetism are Fe, Co, Ni, Gd, Dy, number of alloys of Cu, Al and Mn, and oxides of metals of 1st transition series like CrO_2 , Fe_3O_4 . Ferromagnetic substances have their permeability very much greater than unity.

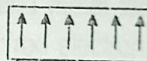


Fig. 25.10 In ferromagnetic substances the neighbouring spins (magnetic moments due to unpaired electrons) align in the same direction).

Effect of Temperature on Susceptibility of Ferromagnetic Substances.

Ferromagnetic behaviour of a substance exists upto a certain temperature which is called Curie temperature or Curie point. This temperature is denoted by T_C . The graph between temperature (T) and susceptibility (χ) of a ferromagnetic substance is given in Figure 25.11. The graph for a paramagnetic substance is also given for the sake of comparison.

It may be seen from graph that below T_C , say at T_1 , χ for a ferromagnetic substance is much higher than that for a normal paramagnetic substance. Ferromagnetic substance does not obey Curie-Weiss law in this range, but above this temperature the ferromagnetic substance obeys Curie-Weiss law.

Do you know that!

A point of distinction between splitting of d-orbitals in tetrahedral and octahedral fields is that g is not used (ٹیٹراہیڈرل میں g استعمال نہیں ہوتا). For example, the orbitals $d_{x^2-y^2}$ and d_{z^2} are designated as e orbitals whereas the other three orbitals d_{xy} , d_{yz} and d_{zx} are designated as t_2 . This is because a tetrahedral geometry has no centre of symmetry (اس ٹیٹراہیڈرل میں ایک جیسا سینٹر نہیں ہوتا). The symbols g and u have relevance (زیر غور معاملہ ہے مربوط ہونے کا عمل) only for fields which have centre of symmetry.

4.7.8 MAGNETIC PROPERTIES OF TRANSITION METAL COMPLEXES:

When the transition metal ions or their complexes are placed in a magnetic field, they show different behaviour. On the basis of this behaviour these substances have been classified as paramagnetic and ferromagnetic.

The transition metal complexes whose central atom/ion contains one or more unpaired electrons are paramagnetic while those with no unpaired electrons behave as diamagnetic substances. The paramagnetic character increases with the increase in the number of unpaired electrons. The paramagnetism of a substance is expressed in terms of its magnetic moment (μ) (مقناطیسی میدان سے حرکت کی پیمائش کا عمل). Larger the number of unpaired electrons in the central atom/ion greater is its paramagnetic character and hence larger is the value of the magnetic movement. The magnetic moment is expressed in Bohr Magnetons (B.M.). Paramagnetic substances have some definite value of magnetic moment while diamagnetic substances have zero value. Ferromagnetism is due to large number of unpaired electrons.

Formula for calculating the magnetic moment of transition metal complexes:

Consider a transition metal complex of 3d-elements whose central metal ion contains only unpaired electrons in its 3d orbitals. The magnetic moment (μ) of an unpaired electron is equal to the sum of its magnetic moment due to its orbital motion (آرٹس میں حرکت پذیر ہوتا), called orbital magnetic moment (μ_l) and that due to its spin motion (سپن کرتے ہوئے حرکت پذیر ہوتا), called spin magnetic moment (μ_s).

We may write

$$\mu = \mu_l + \mu_s \quad \dots (i)$$

But $\mu_l = \sqrt{l(l+1)} \frac{eh}{4\pi mc} \quad \dots (ii)$

and $\mu_s = 2\sqrt{s(s+1)} \frac{eh}{4\pi mc} \quad \dots (iii)$

$$\begin{aligned} \therefore \mu &= \left[\sqrt{l(l+1)} + 2\sqrt{s(s+1)} \right] \frac{eh}{4\pi mc} \\ &= \left[\sqrt{l(l+1)} + 2\sqrt{s(s+1)} \right] \text{B.M.} \quad \dots (iv) \end{aligned}$$

where B.M. is called Bohr Magnetron which is equal to $\frac{eh}{4\pi mc}$. B.M. is the unit of magnetic moment.

One B.M. is equal to 9.27×10^{-21} ergs/gauss

Here e = charge on the electron.

h = Planck's quantum number,

m = mass of the electron and

c = velocity of light.

In case of transition metal ions of 3d-series, the ligands surrounding the metal ions restrict the movement of the electron round the nucleus in orbitals and hence the magnetic moment due to the orbital motion of the electron (μ_l) is quenched (پابند کر لیتے ہیں) or $\sqrt{l(l+1)} = 0$.

Thus the magnetic moment of an unpaired electron residing in a 3d-orbital of the ions of 3d-series elements is given by:

$$\mu = 2\sqrt{s(s+1)} \text{ B.M.} \quad \dots (v)$$

In fact, the same equation is valid (مقبول) for large number of compounds of elements other than 3d elements also.

Value of s for one electron = $\frac{1}{2}$. Substituting for s in the above equation, we get,

$$\mu = 2\sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right)} \text{ B.M.}$$

With n unpaired electrons, $\mu = 2\sqrt{\frac{n}{2}\left(\frac{n}{2}+1\right)} \text{ B.M.}$
 $= \sqrt{n(n+2)} \text{ B.M.}$

$$\begin{aligned} \mu &= 2\sqrt{\frac{n}{2}\left(\frac{n}{2}+1\right)} \text{ B.M.} \\ \mu &= 2\sqrt{\frac{n}{2}\left(\frac{n+2}{2}\right)} \\ \mu &= \sqrt{n(n+2)} \text{ B.M.} \end{aligned}$$

Since the value of μ is only due to the spin motion of n unpaired electrons, it is also called spin-only magnetic moment (جو صرف سپن کی حرکت کی وجہ سے ہے) and is represented as μ_s . Thus:

$$\mu = \mu_s = \sqrt{n(n+2)} \text{ B.M.} \quad (v)$$

Equation (v) shows that the magnitude of μ depends on the number of unpaired electrons (n) present in a given species. The value of μ for different number of unpaired electrons can be calculated with the help of equation (v) as shown below:

For $n=0$ $\mu_s = \sqrt{0(0+2)} = 0.0 \text{ B.M.}$

For $n=1$ $\mu_s = \sqrt{1(1+2)} = \sqrt{3} = 1.73 \text{ B.M.}$

4.7.9 CRYSTAL FIELD THEORY (CFT) AND MAGNETIC PROPERTIES OF COMPLEXES.

CFT has helped us to understand the magnetic properties of coordination compounds. The substances which have all paired electrons are called diamagnetic. The substances which contain unpaired electrons are called paramagnetic substances. Magnitude of Δ_0 and pairing energy P play an important role in this direction.

- (i) If $P > \Delta_0$, the electrons will not pair up and the complex will be high spin complex.
- (ii) If $\Delta_0 > P$, the electrons will prefer to pair up and the complex will be low spin complex.

Thus, the complexes with weak field ligands are high spin complexes and are paramagnetic. Those with strong field ligands are low spin complexes and are diamagnetic.

This generalization could not be provided by valence bond theory. Consider the following two complexes of cobalt (III). From experiments, it has been observed that the complex $[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic while the complex $[\text{CoF}_6]^{3-}$ is paramagnetic.

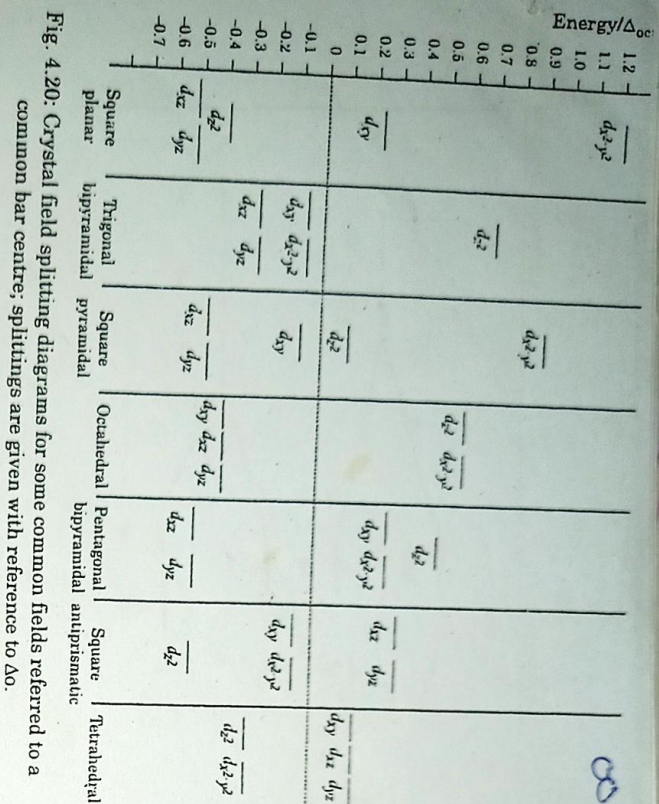


Fig. 4.20: Crystal field splitting diagrams for some common fields referred to a common bar centre; splittings are given with reference to Δ_o .

Limitations of Crystal Field Theory

The crystal field model is successful in explaining the formation, structures, colour and magnetic properties of coordination compounds to a large extent. However, from the assumptions that the ligands are point charges, it follows that anionic ligands should exert the greatest splitting effect. The anionic ligands actually are found at the low end of the spectrochemical series. Further, it does not take into account the covalent character of bonding between the ligand and the central atom. These are some of the weaknesses of CFT, which are explained by ligand field theory (LFT) and molecular orbital theory which are beyond the scope of the present study.

4.10 JAHN-TELLER DISTORTIONS

The Jahn-Teller effect is a geometric distortion of a non-linear molecular system that reduces its symmetry and energy. This distortion is typically observed among octahedral complexes where the two axial bonds can be shorter or longer than those of the equatorial bonds. This effect can also be observed in tetrahedral compounds. This effect is dependent on the electronic state of the system. In 1937, Hermann Jahn and Edward Teller postulated a theorem stating that "stability and degeneracy are not possible simultaneously unless the molecule is a linear one," in regards to its electronic state. This leads to a break in degeneracy which stabilizes the molecule and by consequence, reduces its symmetry. Since 1937, the theorem has

been revised which Husecroft and Sharpe have eloquently phrased as "any non linear molecular system in a degenerate electronic state will be unstable and will undergo distortion to form a system of lower symmetry and lower energy, thereby removing the degeneracy." This is most commonly observed with transition metal octahedral complexes, however, it can be observed in tetrahedral compounds as well.

For a given octahedral complex, the five d atomic orbitals are split into two degenerate sets when constructing a molecular orbital diagram. These are represented by the sets symmetry labels: t_{2g} (d_{xy}, d_{xz}, d_{yz}) and e_g ($d_{x^2-y^2}$ and d_{z^2}). When a molecule possesses a degenerate electronic ground state, it will distort (Jahn-Teller effect) to remove the degeneracy and form a lower energy (and by consequence, lower symmetry) system. The octahedral complex will either elongate or compress the z ligand bonds as shown in Fig. 4.21 below:

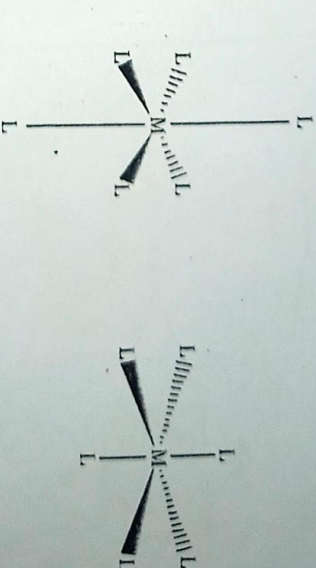


Fig. 4.21: Jahn-Teller distortions for an octahedral complex.

When an octahedral complex exhibits elongation, the axial bonds are longer than the equatorial bonds. For a compression, it is the reverse; the equatorial bonds are longer than the axial bonds. Elongation and compression effects are dictated by the amount of overlap between the metal and ligand orbitals. Thus, this distortion varies greatly depending on the type of metal and ligands. In general, the stronger the metal-ligand orbital interactions are, the greater the chance for a Jahn-Teller effect to be observed.

Elongation

Elongation in Jahn-Teller distortions occur when the degeneracy is broken by the stabilization (lowering in energy) of the d orbitals with a z component, while the orbitals without a z component are destabilized (higher in energy) as shown in Fig. 4.22 below:

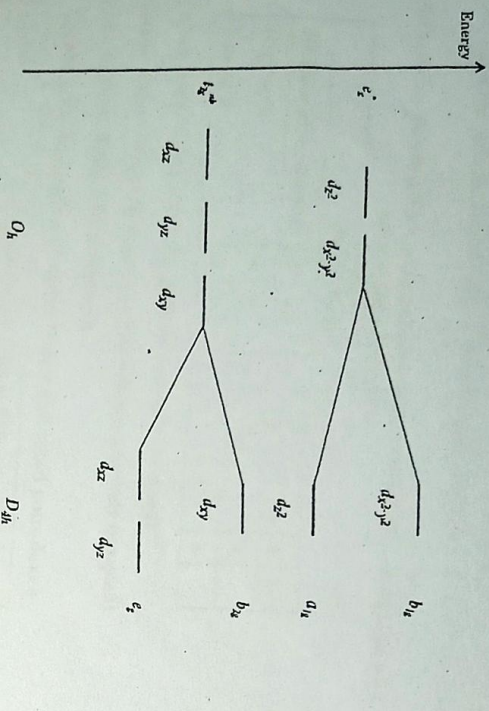


Fig.4.22: Illustration of tetragonal distortion (elongation) for an octahedral complex.

This is due to the d_{xy} and $d_{x^2-y^2}$ orbitals having greater overlap with the ligand orbitals, resulting in the orbitals being higher in energy. Since the $d_{x^2-y^2}$ orbital is antibonding, it is expected to increase in energy due to elongation. The d_{xy} orbital is still nonbonding, but is destabilized due to the interactions. Jahn-Teller elongations are well-documented for copper(II) octahedral compounds. A classic example is that of copper(II) fluoride as shown in Fig.4.23; below:

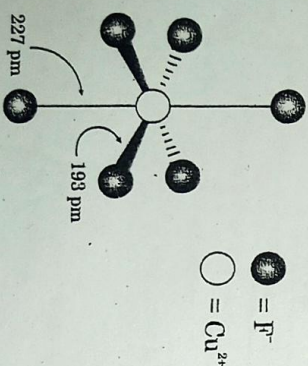


Fig.4.23: Structure of octahedral copper(II) fluoride.

Notice that the two axial bonds are both elongated and the four shorter equatorial bonds are the same length as each other. According to the theorem, the orbital degeneracy is eliminated by distortion, making the molecule more stable based on the model presented in Figure 2.

Compression

Compression in Jahn-Teller distortions occur when the degeneracy is broken by the stabilization (lowering in energy) of the d orbitals without a z component, while the orbitals with a z component are destabilized (higher in energy) as shown in Fig.4.26 below:

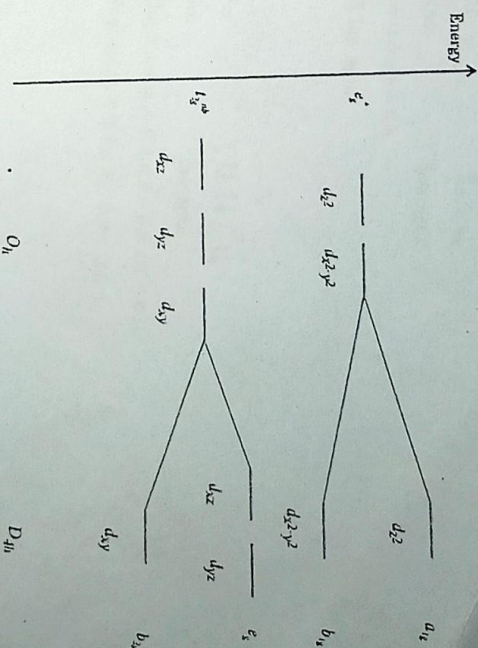


Fig.4.24: Illustration of tetragonal distortion (compression) for an octahedral complex.

This is due to the z -component d orbitals having greater overlap with the ligand orbitals, resulting in the orbitals being higher in energy. Since the $d_{x^2-y^2}$ orbital is antibonding, it is expected to increase in energy due to compression. The d_{xz} and d_{yz} orbitals are still nonbonding, but are destabilized due to the interactions.

Spin States

For Jahn-Teller effects to occur in transition metals there must be degeneracy in either the t_{2g} or e_g orbitals. The electronic states of octahedral complexes are classified as either low spin or high spin. The spin of the system is dictated by the chemical environment. This includes the characteristics of the metal center and the types of ligands.

(i) Low Spin

Fig.4.25 (below) shows the various electronic configurations for low spin octahedral complexes:

The expected effects for octahedral coordination are given in the following table:

Table 4.13: Jahn-Teller effect with reference to number of d electrons and spin state of complex

Jahn-Teller effect										
Number of d electrons	1	2	3	4	5	6	7	8	9	10
High spin	w	w		s		w	w		s	
Low spin	w	w		w		w		s		s

w: weak Jahn-Teller effect (t_{2g} orbitals unevenly occupied)

s: strong Jahn-Teller effect expected (e_g orbitals unevenly occupied)
blank: no Jahn-Teller effect expected.

Table 4.14: Examples of Jahn-Teller distorted complexes

CuBr ₂	4 Br at 240pm 2 Br at 318pm
CuCl ₂	4 Cl at 230pm 2 Cl at 295pm
CuCl ₂ ·2H ₂ O	2 O at 193pm 2 Cl at 228pm 2 Cl at 295pm
CsCuCl ₃	4 Cl at 230pm 2 Cl at 265pm
CuF ₂	4 F at 193pm 2 F at 227pm
CuSO ₄ ·4NH ₃ ·H ₂ O	4 N at 205pm 1 O at 253pm 1 O at 337pm
K ₂ CuF ₄	4 F at 191pm 2 F at 237pm
KCuAlF ₆	2 F at 189pm 4 F at 220pm
CrF ₂	4 F at 200pm 2 F at 243pm
KCrF ₃	4 F at 214pm 2 F at 200pm
MnF ₃	2 F at 209pm 2 F at 191pm 2 F at 179pm

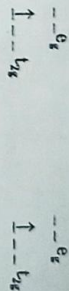
Example 4.1

Calculate the crystal field stabilization energy (CFSE) for d^1 , d^5 and d^7 ions configuration for low and high spin octahedral complexes.

Solution

The crystal field stabilization energies for metal ions in octahedral complexes is calculated by assigning a value of 0.4Δ_o for each electron present in t_{2g} orbitals and -0.6Δ_o for each electron present in e_g orbitals.

1. d^1 ion

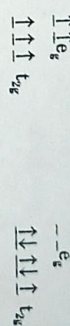


high spin low spin

$$\text{CFSE} = 0.4 \times 1 = 0.4$$

(high & low spin)

2. d^2 ion

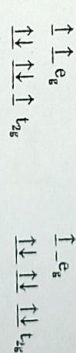


high spin low spin

$$\text{CFSE (high spin)} = 0.4 \times 3 - 0.6 \times 2 = 0.0$$

$$\text{CFSE (low spin)} = 0.4 \times 5 - 0.6 \times 2 = 2.0$$

3. d^7 ion



high spin low spin

$$\text{CFSE (high spin)} = 0.4 \times 5 - 0.6 \times 2 = 0.8$$

$$\text{CFSE (low spin)} = 0.4 \times 6 - 0.6 \times 1 = 1.8$$

Example 11.2

Which complex of the following pairs has the larger value of Δ_o?

- [Co(CN)₆]³⁻ and [Co(NH₃)₆]³⁺
- [Co(NH₃)₆]³⁺ and [CoF₆]³⁻
- [Co(OH₂)₆]²⁺ and [Co(OH₂)₆]³⁺

Solution

Since CN⁻ is stronger ligand than NH₃ (i.e., CN⁻ ion has greater field strength than NH₃), Δ_o for [Co(CN)₆]³⁻ is greater than Δ_o for [Co(NH₃)₆]³⁺.

(ii) [Co(NH₃)₆]³⁺ ion has greater value of Δ_o than that of [CoF₆]³⁻.

(iii) The charge on Co in [Co(OH₂)₆]³⁺ is higher than that on Co in [Co(OH₂)₆]²⁺. Thus Δ_o for [Co(OH₂)₆]³⁺ is higher.

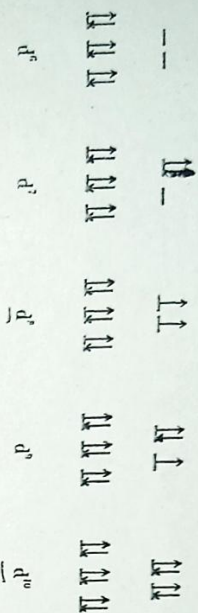


Fig. 4.25: Low spin octahedral coordination diagram (red indicates no degeneracies possible, thus no Jahn-Teller effects).

The figure illustrates that low spin complexes with d^3 , d^8 , d^9 , and d^{10} electrons cannot have Jahn-Teller distortions. These electronic configurations correspond to a variety of transition metals. Many common examples include Cr^{3+} , Co^{3+} , Ni^{2+} , and Cu^{2+} .

(ii) High Spin

Fig. 4.26 below shows the various electronic configurations for high spin octahedral complexes:

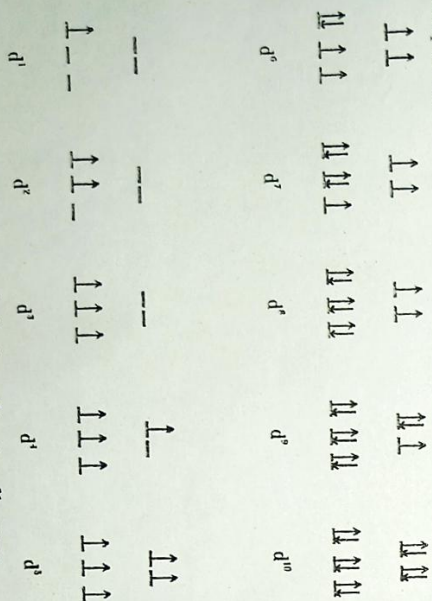


Fig. 4.26: High spin octahedral coordination diagram (red indicates no degeneracies possible, thus no Jahn-Teller effects).

The figure illustrates that low spin complexes with d^1 , d^5 , d^8 , and d^{10} electrons cannot have Jahn-Teller distortions. In general, degenerate electronic states occupying the e_g orbital set tend to show stronger Jahn-Teller effects. This is primarily caused by the occupation of these high energy orbitals. Since the system is more stable with a lower energy configuration, the degeneracy of the e_g set is broken, the symmetry is reduced, and occupations at lower energy orbitals occur.

Spectroscopy

Jahn-Teller distortions can be observed using a variety of spectroscopic techniques. In UV-VIS absorption spectroscopy, distortion causes splitting of bands in the spectrum due to a reduction in symmetry (O_h to D_{4h}). Consider a hypothetical molecule with octahedral symmetry showing a single absorption band. If the molecule were to undergo Jahn-Teller distortion, the number of bands would increase as shown in Fig. 4.27 below:

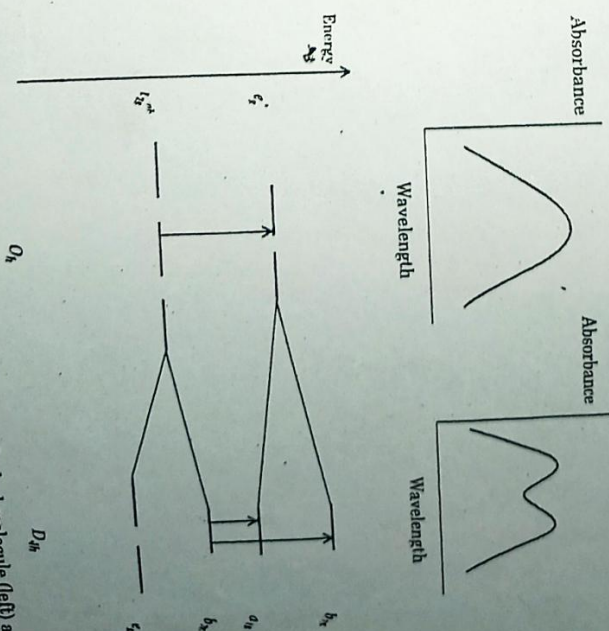


Fig. 4.27: Hypothetical absorption spectra of an octahedral molecule (left) and the same molecule with Jahn-Teller elongation (right). The arrows indicate electronic transitions.

A similar phenomenon can be seen with IR and Raman vibrational spectroscopy. The number of vibrational modes for a molecule can be calculated using the $3n - 6$ rule (or $3n - 5$ for linear geometry) rule. If a molecule exhibits an O_h symmetry point group, it will have fewer bands than that of a Jahn-Teller distorted molecule with D_{4h} symmetry. Thus, one could observe Jahn-Teller effects through either IR or Raman techniques. This effect can also be observed in EPR experiments as long as there is at least one unpaired electron.